



Methods in Differential Equations · Worksheet 1

Core Pure 2 Chapter 7: Methods in Differential Equations. Answer all questions. Show your working clearly. Total: 67 marks. Time: 85 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 7. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Use A , B or c for arbitrary constants. Give general solutions in the form $y = f(x)$ where possible, and give exact answers.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A First-order equations: the integrating factor method	20	
B Second-order homogeneous equations	20	
C Non-homogeneous equations and particular solutions	22	
Roar-level challenge	5	
Total	67	

Section A First-order equations: the integrating factor method

20 marks

1. By separating the variables, find the general solution of the differential equation [2]

$$\frac{dy}{dx} = xy, \quad y > 0.$$

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2. Use an integrating factor to find the general solution of

[3]

$$\frac{dy}{dx} + 2y = e^x.$$

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3. Find the general solution of the differential equation

[3]

$$x \frac{dy}{dx} + 2y = x^2, \quad x > 0.$$

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4. Find the particular solution of the differential equation

[4]

$$\frac{dy}{dx} + y \tan x = \cos x, \quad -\frac{\pi}{2} < x < \frac{\pi}{2},$$

for which $y = 2$ when $x = 0$.

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5. (a) Find the general solution of the differential equation

[5]

$$(x^2 + 1) \frac{dy}{dx} + 4xy = x.$$

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(b) Find the particular solution for which $y = 0$ when $x = 0$.

[2]

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(c) Describe the behaviour of this particular solution as $x \rightarrow \infty$.

[1]

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(Total for Question 5 is 8 marks)



Section B Second-order homogeneous equations

20 marks

6. Find the general solution of

[2]

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0.$$

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7. Find the general solution of

[3]

$$4\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + y = 0.$$

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8. Find the general solution of

[3]

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 0.$$

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9. Find the solution of the differential equation

[4]

$$\frac{d^2y}{dx^2} + 9y = 0$$

for which $y = 2$ and $\frac{dy}{dx} = 6$ when $x = 0$.

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10. (a) Find the general solution of the differential equation

[3]

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0.$$

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(b) Given that $y = 0$ and $\frac{dy}{dx} = 1$ when $x = 0$, find the particular solution.

[3]

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(c) Find the smallest positive value of x at which y has a stationary point.

[2]

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(Total for Question 10 is 8 marks)



Section C Non-homogeneous equations and particular solutions

22 marks

11. Find a particular integral of

[2]

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 4.$$

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12. Find a particular integral of

[3]

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 2y = 4x.$$

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13. Find the general solution of

[3]

$$\frac{d^2y}{dx^2} + 4y = 5e^x.$$

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14. Find the general solution of

[4]

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 3e^{2x}.$$

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15. The differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 10\sin x$$

describes a physical system.

(a) Find the complementary function. [2]

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(b) Given that there is a particular integral of the form $y = a\cos x + b\sin x$, find the values of a and b . [3]

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(c) Hence find the particular solution for which $y = 0$ and $\frac{dy}{dx} = 0$ when $x = 0$. [4]

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(d) Describe the behaviour of y for large values of x . [1]

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(Total for Question 15 is 10 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. Find the general solution of the differential equation

[5]

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = e^{2x}.$$

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END OF WORKSHEET

Total: 67 marks